

Technical Report: Harbor Site Selection

Project: İğneada Commercial Hub

Group 3

1 Introduction

The selection of İğneada, located in the Kırklareli province, is based on the strategic necessity of expanding Turkey's Black Sea trade capacity. A harbor is defined as a coastal installation providing shelter and maintenance for ships, while a port includes terminal and transfer facilities such as piers, warehouses, and cargo handling equipment. This project is classified as a "Commercial Harbor" intended to function as a "Hub" for international maritime traffic.

2 Region Selection (Foreland and Hinterland)

2.1 Foreland and Transshipment Potential

İğneada is positioned near a vital international seaway leading to the Bosphorus. Its proximity to European maritime borders makes it a prime candidate for transshipment operations, specifically involving the Black Sea loop with countries such as Bulgaria, Romania, and Russia. The site sits at the intersection of a major seaway and the projected land-based highway networks.

2.2 Hinterland Analysis

The inland area (Hinterland) consists of the Thrace region, which offers:

- **Economic Drivers:** Developing cities with diverse industrial and agricultural products ready for export.
- **Connectivity:** Topography that is not excessively high or hilly, which facilitates the construction of efficient railway and highway connections to Istanbul and Europe.

3 Local Site Investigation

3.1 Waves and Hydrodynamics

As an "open sea harbor," İğneada is directly exposed to the predominant wave direction of the North Black Sea. This exposure necessitates the design and construction of robust,

long breakwaters to create an artificial sheltered basin. This provides a target environment for rubble mound or pile-based breakwater designs similar to those implemented in the Filyos or Çandarlı port projects.

3.2 Bathymetry and Water Depth

A detailed bathymetric map is required to ensure the site involves the least amount of dredging. The design must balance two cost heavy factors: if the water is too shallow, dredging becomes prohibitively expensive; if it is too deep, the volume and cost of material for breakwater construction increase significantly.

3.3 Geotechnical Conditions

Favorable soil conditions are essential for the heavy infrastructure of a port terminal. Our site selection avoids areas dominated by soft clay, which would require expensive soil improvement or deep piling to reach bedrock. We must also investigate the presence of underwater rock, as underwater excavation is a high-cost operation that should be avoided where possible.

4 Traffic Forecast and Facility Requirements

4.1 Facility Specifications

The transition from a harbor to a port involves the integration of:

- **Marine Infrastructure:** Navigation channels, anchorage areas, and mooring basins.
- **Terminal Infrastructure:** Cargo handling equipment, specialized terminals (General Cargo, Oil, or Dry Bulk), and extensive warehousing.

5 Environmental and National Policy

Site selection is further influenced by non technical factors such as national defense policies and political strategic goals. Environmental impact assessments must focus on the aesthetic preservation of the coastline and the management of ship generated waste to prevent pollution of the local marine ecosystem.

6 Conclusion

İğneada stands as the optimal choice for a new maritime hub between Kırklareli and Zonguldak. It satisfies the technical requirements for hinterland accessibility and transshipment potential while presenting the necessary maritime challenges specifically wave exposure to justify and implement advanced breakwater engineering.